Efficient Control Design in Open Quantum Systems under Parametric Uncertainty

Yidian Fan and Re-Bing Wu

Department of Automation, Tsinghua University

fanyd23@mails.tsinghua.edu.cn

IEEE Conference on Decision and Control 2025

CDC25_0269 Invited Session Paper Presentation

Motivation

➢Uncertainty can be prevalent in quantum control systems

☐ Quantum computation on superconducting qubits

- parasitic coupling
- TLS defects
- spurious microwave modes
- coupling to control lines
- electronics noise
- pulse distortions
- microwave-carrier bleedthrough

☐ NMR system

- Larmor dispersion
- rf-inhomogenity
- variations of couplings

➢Uncertain quantum systems

unknown strength

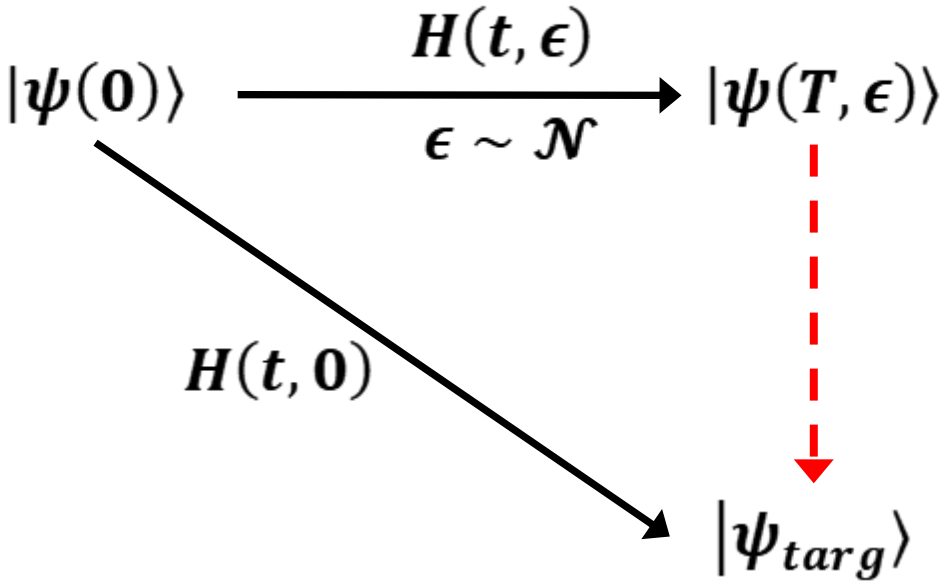
$$H(t,\epsilon) = H_0 + \sum_c u_c(t)H_c + \epsilon E$$

- Frequency-shift $E = \sigma_z$
- ZZ-Crosstalk $E = \sigma_z^{(1)}\sigma_z^{(2)}$
- Coupling fluctuations

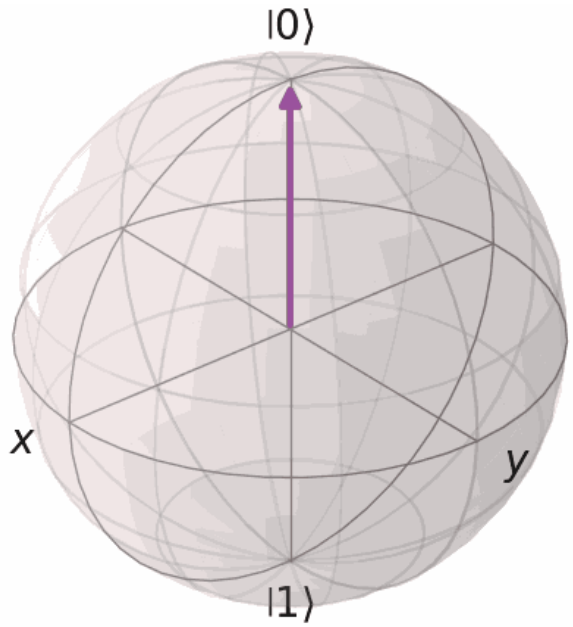
$$E = \sigma_x^{(1)}\sigma_x^{(2)} + \sigma_y^{(1)}\sigma_y^{(2)}$$

Motivation

➢Robustness



Robust control: Steer an **ensemble** of trajectories originating from the same initial state to the same target state



Introduction

➢ Expansion-based robust control design

Taylor expand: $|\psi(T,\epsilon)\rangle = |\psi(T,0)\rangle + \epsilon|\psi_1\rangle + \frac{\epsilon^2}{2}|\psi_2\rangle + \ldots$

Truncation

$$\max_{u(t)} J = \left|\langle\psi_{targ}|\psi(T,0)\rangle\right|^2 - \sum_{i=1}^{n_r} \lambda_i \||\psi_i\rangle\|^2$$

- Explicit and Quantifiable Suppression of Error Sensitivity
- Flexible and Tunable Fidelity-Robustness Tradeoff
- Accelerated Convergence via Quasi-Newton Optimizers

Introduction

➢High computational cost

$$J = \left|\langle \psi_{targ} | \psi(T,0) \rangle\right|^2 - \sum_{i=1}^{n_r} \lambda_i \| |\psi_i\rangle \|^2$$

Augmented system:

$$\vec{\psi} = \begin{bmatrix} \psi_{n_r} \\ \vdots \\ \psi_2 \\ \psi_1 \\ \psi_0 \end{bmatrix} \in \mathbb{C}^{n_r d \times d}$$

$$\dot{\vec{\psi}}(t) = A[u(t)]\vec{\psi}(t)$$

- Deterministic system
- Higher system dimension

- The system evolution becomes computationally expensive when the robustness order or the system dimension is modestly high
- This cost grows even more severe when dealing with the open-system dynamics

Outline

- System Formulation
- Suzuki-Trotter Expansion
- Experimental Results
- Concluding Remarks

Uncertain open quantum system

Consider an uncertain open quantum system with the following dynamics:

$$\dot{\rho}(t) = -i[H(t,\epsilon),\rho(t)] + \sum_{i=1}^{n_d} \gamma_i \left[c_i \rho(t) c_i^\dagger - \frac{1}{2} c_i^\dagger c_i \rho(t) - \frac{1}{2} \rho(t) c_i^\dagger c_i\right]$$

- the system Hamiltonian: $H(t,\epsilon) = H_S(t) + \sum_{j=1}^{m} \epsilon_j E_j$
- $H_S(t) = H_0 + \sum_{c=1}^{n_c} u_c(t) H_c$, $\rho(t)$ is the density matrix, and c_i are the Lindblad operators that model incoherent dynamics with rate γ_i, typically arising from environmental interactions

We explicitly single out the effects of uncertainty in a *superoperator* formalism

$$\dot{\rho}(t) = \left[\mathcal{L}(t) + \sum_{j=1}^{m} \epsilon_j \mathcal{E}_j\right] \rho(t)$$

- where $\mathcal{L}(t)\rho(t) := -i[H_s(t),\rho(t)] + \sum_{i=1}^{n_d} \gamma_i \left[c_i \rho(t) c_i^\dagger - \frac{1}{2} c_i^\dagger c_i \rho(t) - \frac{1}{2} \rho(t) c_i^\dagger c_i\right]$

$$\mathcal{E}_j \rho(t) := -i[E_j, \rho(t)]$$

Augmented system

- **Taylor expand:** $\rho(T) = \sum_{p_1,\cdots,p_m \geq 0} \epsilon_1^{p_1} \cdots \epsilon_m^{p_m} \rho_{p_1,\cdots,p_m}(T)$

$$\dot{\rho}_0(t) = \mathcal{L}(t)\rho_0(t) \qquad \dot{\rho}_{p_1,\cdots,p_m}(t) = \mathcal{L}(t)\rho_{p_1,\cdots,p_m}(t) + \sum_{\substack{j=1 \\ p_j \geq 1}}^{m} \mathcal{E}_j \rho_{p_1,\cdots,p_j-1,\cdots,p_m}(t)$$

- **Augmented system formulation:** Truncate to finite terms with $0 \leq p_1 + \cdots + p_m \leq n$

$$\vec{\rho}(t) = \begin{bmatrix} \rho_{n,0,\cdots,0}(t) \\ \vdots \\ \rho_{p_1,\cdots,p_m}(t) \\ \vdots \\ \rho_{0,\cdots,0,1}(t) \\ \rho_0(t) \end{bmatrix}$$

$$\dot{\vec{\rho}}(t) = \mathbf{L}(t)\vec{\rho}(t) + \sum_{j=1}^{m} \mathbf{E}_j \vec{\rho}(t)$$

- $\mathbf{L}(t) = \begin{bmatrix} \mathcal{L}(t) & 0 & \cdots & 0 \\ 0 & \mathcal{L}(t) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathcal{L}(t) \end{bmatrix}$
- $\mathbf{E}_j = R_j \otimes \mathcal{E}_j$

$$\begin{bmatrix} \vdots \\ \dot{\rho}_{\cdots,p_j,\cdots}(t) \\ \vdots \\ \vdots \\ \vdots \end{bmatrix} = \begin{bmatrix} \mathcal{L}(t) & & & & \\ & \mathcal{L}(t) & & & \\ & & \mathcal{L}(t) & & \\ & & & \ddots & \\ & & & & \mathcal{L}(t) \end{bmatrix} \begin{bmatrix} \vdots \\ \rho_{\cdots,p_j,\cdots}(t) \\ \vdots \\ \vdots \\ \vdots \end{bmatrix} + \sum_j \begin{bmatrix} 0 & & & & \\ & 0 & \cdots & \mathcal{E}_j & \cdots \\ & & 0 & & \\ & & & \ddots & \\ & & & & 0 \end{bmatrix} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \rho_{\cdots,p_j-1,\cdots}(t) \\ \vdots \end{bmatrix}$$

Gradient-based optimization

☐ **Robust control objective function:**

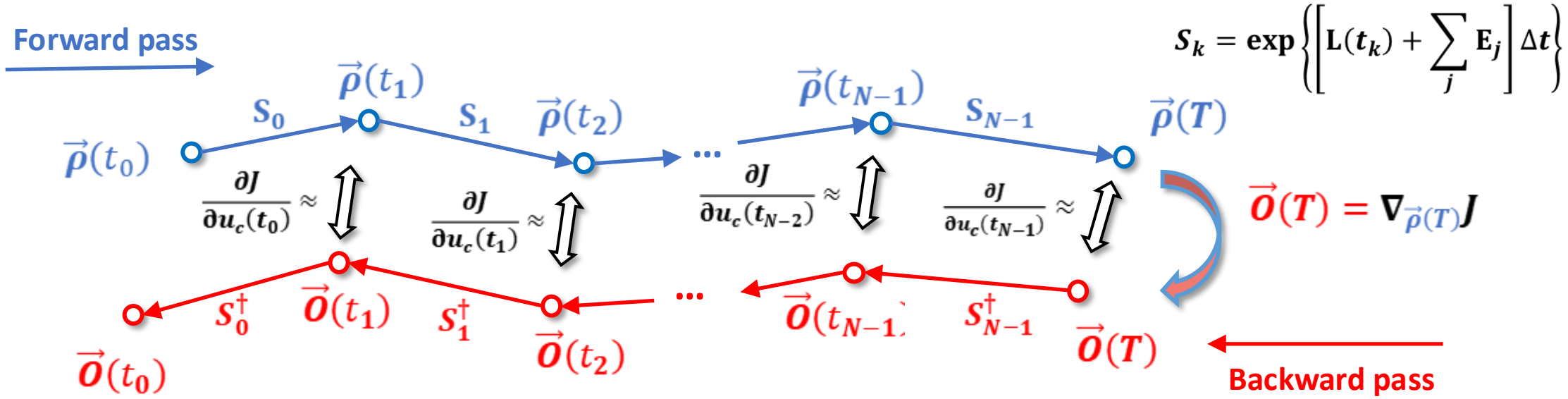
$$J[\vec{\rho}(T)] = \langle \psi_{\text{targ}} | \rho_0(T) | \psi_{\text{targ}} \rangle - \frac{1}{2} \sum_{0<p_1+\cdots+p_m\leq n} \lambda_{p_1,\cdots,p_m} \| \rho_{p_1,\cdots,p_m}(T) \|_F^2$$

- **Consider piece-wise constant control fields** $\{u_c(t_k)\}$, $t_k = k\Delta t$, $\Delta t = t/N$

$$\frac{\partial J}{\partial u_c(t_k)} = \left\langle \vec{O}(T), \frac{\partial \vec{\rho}(T)}{\partial u_c(t_k)} \right\rangle \qquad \vec{O}(T) = \nabla_{\vec{\rho}(T)} J = \begin{bmatrix} \vdots \\ -\lambda_{p_1,\cdots,p_m} \rho_{p_1,\cdots,p_m}(T) \\ \vdots \\ -\lambda_{0,\cdots,0,1} \rho_{0,\cdots,0,1}(T) \\ |\psi_{targ}\rangle\langle\psi_{targ}| \end{bmatrix}$$

- **Follow the GRAPE algorithm to perform gradient-based control update**

GRAPE



- For efficiency, the gradient $\frac{\partial J}{\partial u_c(t_k)}$ is typically approximated with $\mathcal{O}(\Delta t^2)$ precision
- The gradient-based control update can then be performed using the approximated gradient
- Most of the computational cost arises from evaluating the forward and backward dynamics

Computational cost

- **Matrix Representation:**

$$S_k = \exp\left\{\left[\mathbf{L}(t_k) + \sum \mathbf{E}_j\right]\Delta t\right\}$$

Suppose the dimension of the quantum system is d, and $\vec{\rho}(t)$ consists of M blocks

$$\rho = \begin{bmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{bmatrix} \xrightarrow{\text{vectorization}} \begin{bmatrix} \rho_{11} \\ \rho_{21} \\ \rho_{12} \\ \rho_{22} \end{bmatrix}$$

	Size after vectorization
$\rho(t)$	$d^2 \times 1$
$\vec{\rho}(t)$	$Md^2 \times 1$
$\mathbf{L}(t)$	$Md^2 \times Md^2$
$\mathbf{E}_j$	$Md^2 \times Md^2$

- The computational complexity of evaluating S_k by direct matrix exponentiation is $\mathcal{O}(M^3 d^6)$
- Using an ODE Solver to evolve the augmented system reduces the complexity to $\mathcal{O}(M^2 d^4)$

Outline

- System Formulation
- Suzuki-Trotter Expansion
- Experimental Results
- Concluding Remarks

Suzuki-Trotter Expansion

- **Suzuki-Trotter expansion is a widely applied method for approximating the evolution of quantum systems**
  - **First-order S-T expansion:** $$e^{\sum_{j=1}^{m} L_j \Delta t} = \prod_{j=1}^{m} e^{L_j \Delta t} + \mathcal{O}(\Delta t^2)$$
  - **Second-order S-T expansion:** $$e^{\sum_{j=1}^{m} L_j \Delta t} = \prod_{j=1}^{m} e^{\frac{L_j \Delta t}{2}} \prod_{j=m}^{1} e^{\frac{L_j \Delta t}{2}} + \mathcal{O}(\Delta t^3)$$
- **We generalize approach proposed in , which was originally developed for control optimization of closed systems, to the augmented system formulated for robust control design**

Suzuki-Trotter Expansion

➢ **Suzuki–Trotter expansion can accelerate the evolution of the augmented system**

- For simplicity, assume there is only one diagonal control Hamiltonian H_1
- The open system evolution $\mathcal{L}(t)$ can be decomposed as

$$\mathcal{L}(t) = \mathcal{H}_{eff} + u_1(t)\mathcal{H}_1 + \mathcal{C}, \quad \text{where } \mathcal{H}_{eff}\rho(t) := -i\left[H_{eff}\rho(t) - \rho(t)H_{eff}^{\dagger}\right], \ \mathcal{H}_1\rho(t) := -i[H_1, \rho(t)],$$
$$\mathcal{C}\rho(t) := \sum_i \gamma_i c_i \rho(t) c_i^{\dagger}$$

Suzuki-Trotter expansion

$$S_k = \exp\left\{\left[\mathrm{H_{eff}} + u_1(t_k)\mathrm{H}_1 + \mathrm{C} + \sum_{j=1}^{m}\mathrm{E_j}\right]\Delta t\right\}$$
$$\approx \left(\prod_{j=1}^{m} e^{\mathrm{E_j}\frac{\Delta t}{2}}\right) e^{\mathrm{C}\frac{\Delta t}{2}} e^{u_1(t_k)\mathrm{H}_1\frac{\Delta t}{2}} e^{\mathrm{H_{eff}}\Delta t} e^{u_1(t_k)\mathrm{H}_1\frac{\Delta t}{2}} e^{\mathrm{C}\frac{\Delta t}{2}} \left(\prod_{j=1}^{m} e^{\mathrm{E_j}\frac{\Delta t}{2}}\right) + \mathcal{O}(\Delta t^3)$$

Back

Suzuki-Trotter Expansion

➢ $e^{H_{eff}\Delta t}$ and $e^{u_1(t_k)H_1\frac{\Delta t}{2}}$

- Let $U_{\text{eff}} = \exp(-iH_{\text{eff}}\Delta t)$, $U_1(t_k) = exp\left(-iu_1(t_k)H_1\frac{\Delta t}{2}\right)$

$$e^{\mathrm{H_{eff}}\Delta t}\begin{bmatrix}\rho_{n,0,\cdots,0}(t)\\ \vdots\\ \rho_{p_1,\cdots,p_m}(t)\\ \vdots\\ \rho_{0,\cdots,0,1}(t)\\ \rho_0(t)\end{bmatrix} = \begin{bmatrix}U_{eff}\ \rho_{n,0,\cdots,0}(t)\ U_{eff}{}^{\dagger}\\ \vdots\\ U_{eff}\ \rho_{p_1,\cdots,p_m}(t)\ U_{eff}{}^{\dagger}\\ \vdots\\ U_{eff}\ \rho_{0,\cdots,0,1}(t)\ U_{eff}{}^{\dagger}\\ U_{eff}\ \rho_0(t)\ U_{eff}{}^{\dagger}\end{bmatrix}$$

$$e^{u_1(t_k)\mathrm{H}_1\frac{\Delta t}{2}}\begin{bmatrix}\rho_{n,0,\cdots,0}(t)\\ \vdots\\ \rho_{p_1,\cdots,p_m}(t)\\ \vdots\\ \rho_{0,\cdots,0,1}(t)\\ \rho_0(t)\end{bmatrix} = \begin{bmatrix}U_1(t_k)\rho_{n,0,\cdots,0}(t)\ U_1{}^{\dagger}(t_k)\\ \vdots\\ U_1(t_k)\rho_{p_1,\cdots,p_m}(t)\ U_1{}^{\dagger}(t_k)\\ \vdots\\ U_1(t_k)\rho_{0,\cdots,0,1}(t)\ U_1{}^{\dagger}(t_k)\\ U_1(t_k)\rho_0(t)\ U_1{}^{\dagger}(t_k)\end{bmatrix}$$

	Computational Complexity
U_{eff}	$\mathcal{O}(d^3)$
$U_1(t_k)$	$\mathcal{O}(d)$
$e^{H_{eff}\Delta t}\vec{\rho}(t_k)$	$\mathcal{O}\left(Md^3\right)$
$e^{u_1(t_k)H_1\frac{\Delta t}{2}}\vec{\rho}(t_k)$	$\mathcal{O}\left(Md^3\right)$

- The computational complexity is reduced to $\mathcal{O}\left(Md^3\right)$
- The propagator U_{eff} needs to be computed only once
- Although $U_1(t_k)$ is time-dependent, the computational cost is minimized when H_1 is diagonal

Suzuki-Trotter Expansion

➢ $e^{C\frac{\Delta t}{2}} = \begin{bmatrix} e^{C\frac{\Delta t}{2}} & 0 & \dots & 0 \\ 0 & e^{C\frac{\Delta t}{2}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{C\frac{\Delta t}{2}} \end{bmatrix}$

- We can solve the matrix representation of $e^{C\frac{\Delta t}{2}}$ once
- Or we can approximate the operation using the Taylor expansion

$$e^{C\frac{\Delta t}{2}}\rho_{p_1,\cdots,p_m}(t) \approx \rho_{p_1,\cdots,p_m}(t) + \frac{\Delta t}{2} C\,\rho_{p_1,\cdots,p_m}(t) + \frac{\Delta t^2}{8} C^2 \rho_{p_1,\cdots,p_m}(t) + \mathcal{O}(\Delta t^3)$$

➢ $e^{\mathrm{E_j}\frac{\Delta t}{2}}$ $\mathrm{E_j} = \begin{bmatrix} 0 & \cdots & \mathcal{E}_j & \cdots & \cdots \\ & 0 & \cdots & \mathcal{E}_j & \cdots \\ & & 0 & \cdots & \mathcal{E}_j \\ & & & \ddots & \vdots \\ & & & & 0 \end{bmatrix}$

- E_j is strictly upper triangular, which is nilpotent: $\mathrm{E}_j^{n+1} = 0$

$$e^{\mathrm{E_j}\frac{\Delta t}{2}} = I + \mathrm{E_j}\frac{\Delta t}{2} + \dots + \frac{1}{n!}\left(\mathrm{E_j}\frac{\Delta t}{2}\right)^n$$

The overall complexity of computing the forward and backward dynamics using the Suzuki–Trotter expansion is $\mathcal{O}\left(Md^3\right)$, which is significantly lower than the $\mathcal{O}(M^3d^6)$ and $\mathcal{O}(M^2d^4)$ complexities of the previous approaches

Optimization Landscape

➢ **Approximation Error**

- $S_k = S_k^{\mathrm{ST}} + \mathcal{O}(\Delta t^3)$
- $\vec{\rho}^{\mathrm{ST}}(T) = S_{N-1}^{\mathrm{ST}} \cdots S_1^{\mathrm{ST}} S_0^{\mathrm{ST}} \vec{\rho}(0)$
- $\vec{\rho}(T) = \vec{\rho}^{ST}(T) + \mathcal{O}(\Delta t^2)$

$$J^{ST} = \langle \psi_{\mathrm{targ}} | \rho_0^{ST}(T) | \psi_{\mathrm{targ}} \rangle - \frac{1}{2} \sum_{0<p_1+\cdots+p_m\le n} \lambda_{p_1,\cdots,p_m} \| \rho_{p_1,\cdots,p_m}^{ST}(T) \|_F^2$$

$\sim$ $\mathcal{O}(\Delta t^2)$; $\sim$ $\mathcal{O}(\Delta t^4)$

- The approximation error of the optimization landscape is $\mathcal{O}(\Delta t^2)$, i.e.,

$$J = J^{ST} + \mathcal{O}(\Delta t^2)$$

➢ **Gradient in the approximate landscape**

Recall the form of S_k^{ST}

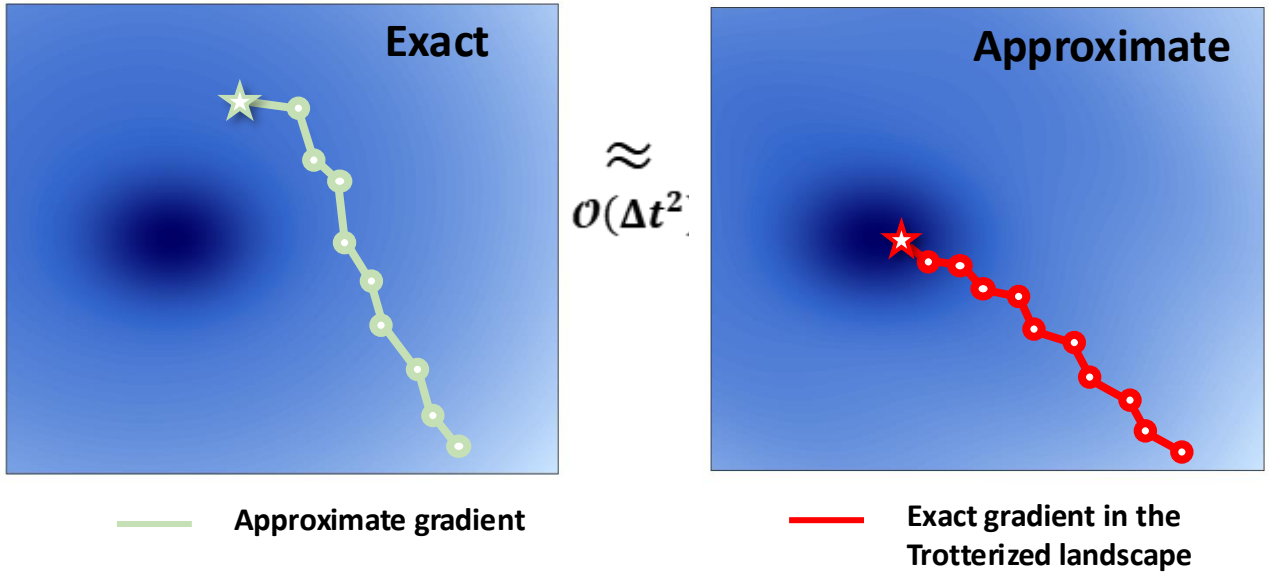
- $\frac{\partial e^{u_1(t_k)\mathrm{H}_1 \frac{\Delta t}{2}}}{\partial u_1(t_k)} = \frac{\Delta t}{2} \mathrm{H}_1 e^{u_1(t_k)\mathrm{H}_1 \frac{\Delta t}{2}}$
- The gradient $\frac{\partial S_k}{\partial u_1(t_k)}$ can then be exactly computed

Trotterization yields an approximate optimization landscape, on which the exact gradient can be computed and utilized

Optimization Landscape

- Jensen's work suggests that in certain optimizers (e.g., BFGS), gradient errors may accumulate over iterations, ultimately leading to degraded performance
- In contrast, within the Trotterized landscape, the availability of exact gradient information offers a greater potential for faithful convergence to its optimum, which can be very close to the true optimum when Δt is sufficiently small

The optimization performance on the Trotterized landscape is both problem- and method-dependent, and for certain control tasks it can outperform traditional approaches



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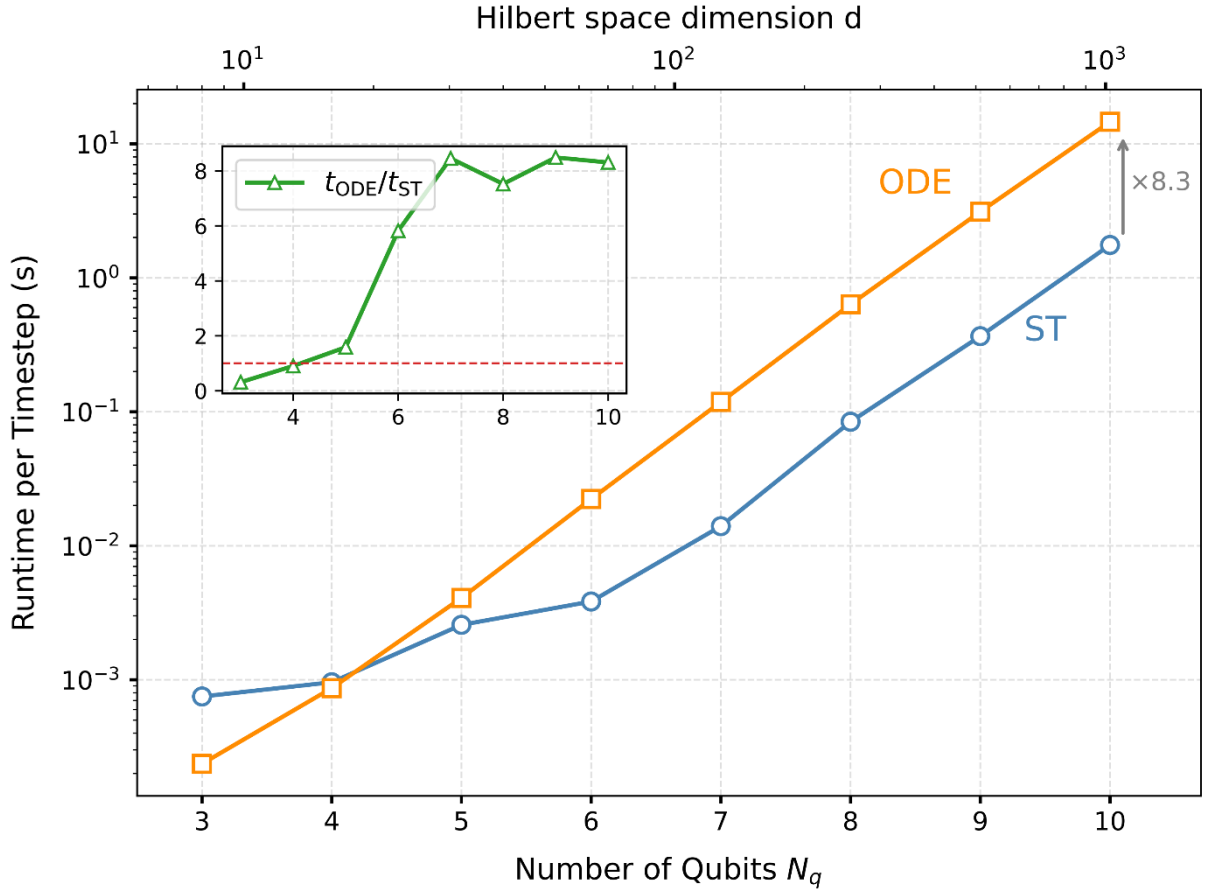
Optimization Landscape

$$H(t) = \sum_{i=1}^{N_q-1} J_{i,i+1}\sigma_i^z\sigma_{i+1}^z + \sum_{i=1}^{N_q}\left[u_{i,x}(t)\sigma_i^x + u_{i,y}(t)\sigma_i^y\right],$$

- where σ_i^α with $\alpha \in \{x, y, z\}$ are the Pauli matrices of the i-th qubit, $J_{i,i+1}/2\pi = 30$ MHz are the strengths of the ZZ interactions of between neighboring qubits
- The Lindblad operators $c_{i,1} = \sigma_i^-$ and $c_{i,2} = \sigma_i^+\sigma_i^-$ represent the dissipation and dephasing effects of the i-th qubit, respectively. The corresponding rates are $\gamma_{i,1} = \frac{1}{T_{i,1}}$ and $\gamma_{i,2} = \frac{1}{T_{i,2}}$, with $T_{i,1} = T_{i,2} = 30\mu s$
- The control pulse is discretized with a timestep of $\Delta t = 0.5$ ns
- The uncertainty terms are $E_1 = \sigma_1^z\sigma_2^z$ and $E_2 = \sigma_{N_q-1}^z\sigma_{N_q}^z$, which models the fluctuations of ZZ-couplings between the first and last pairs of qubits

Experimental Results

- **Choose the robust order $n = 1$. For $3 \leq N_q \leq 10$, the effective dimension of the augmented system $192 \leq Md^2 < 3.15 \times 10^6$**
- **We first compare the average computational time for a single step evolution**



✓ **The S-T expansion accelerates the computation when $N_q \geq 5$, achieving a speedup exceeding 5.8 times for $N_q \geq 6$**

Experimental Results

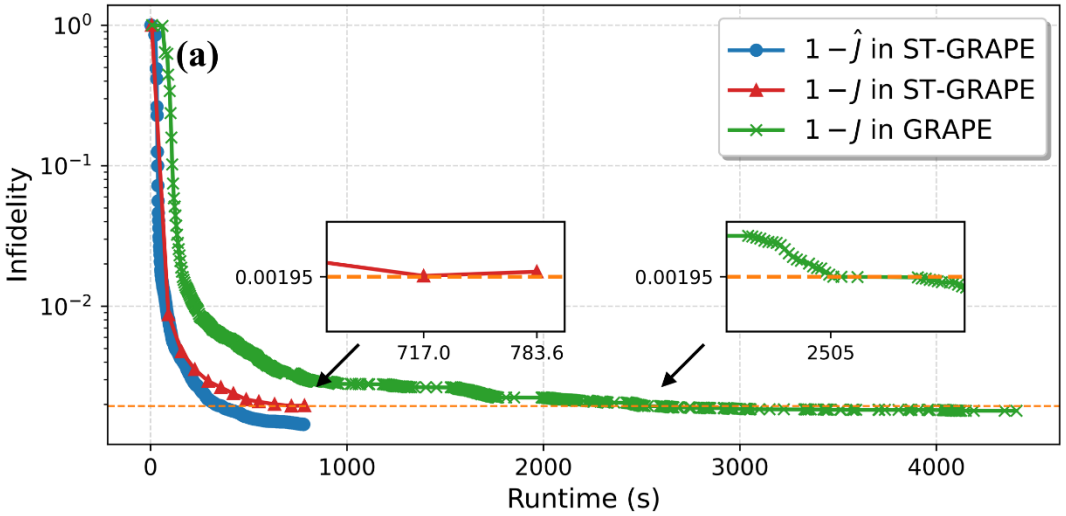
- We next focus on the state preparation task of Hadamard transform, where the initial and target quantum states are

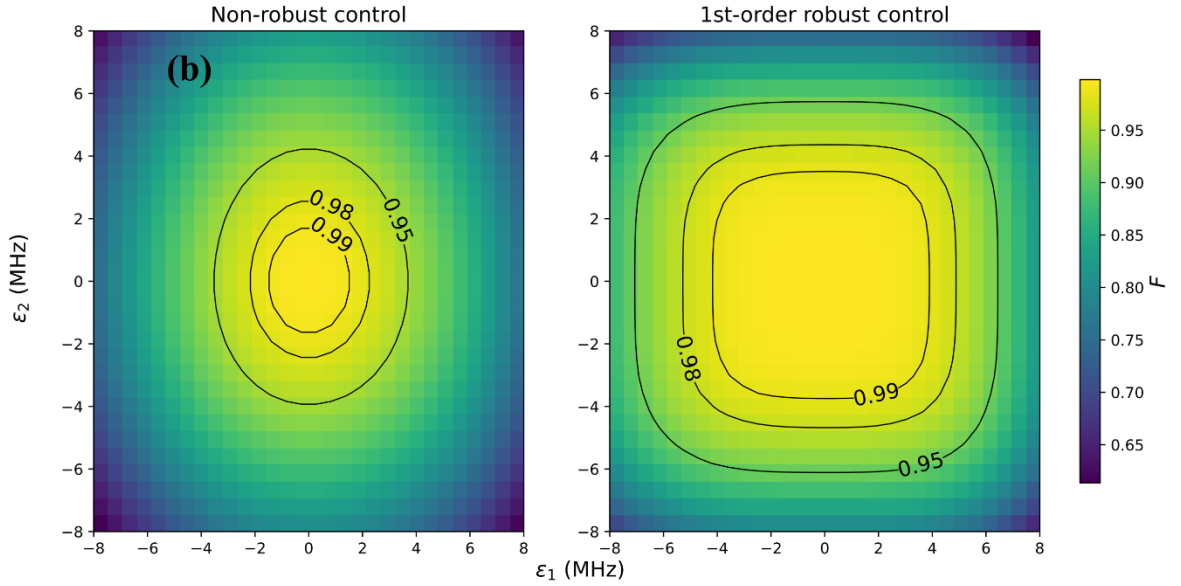
$$|0\rangle^{\otimes N_q} \xrightarrow{\mathrm{H}^{\otimes N_q}} \frac{1}{\sqrt{2^{N_q}}} \sum_{x=0}^{2^{N_q}-1} |x\rangle$$

Here H is the Hadamard gate.

- Let $N_q = 6$, and the pulse duration $T = 40$ ns

✓ A robust solution can be efficiently obtained within a shorter optimization time

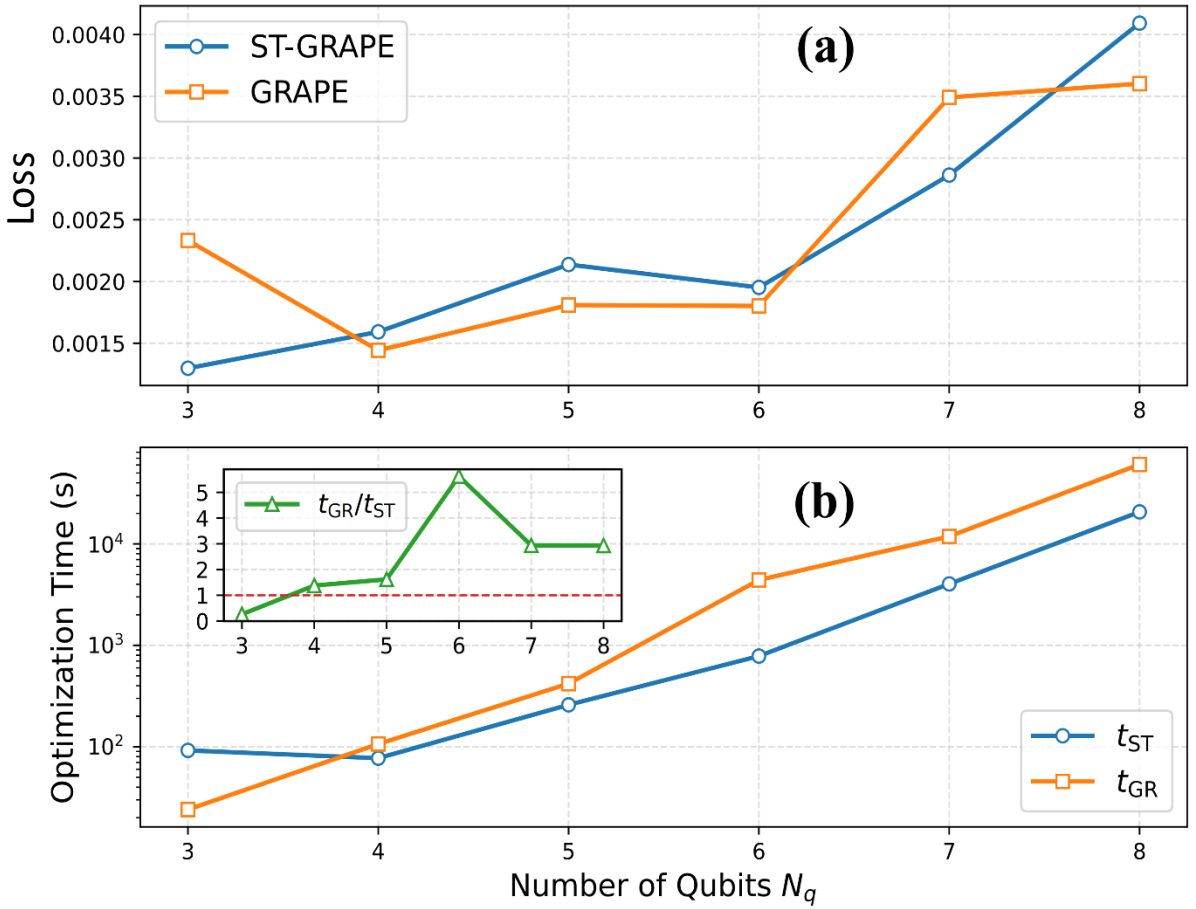




Experimental Results

- **Finally, we evaluate and compare the optimization performance of our proposed algorithm and GRAPE across spin chains of different lengths**

Both algorithms achieve comparable fidelities across all experiments. Our algorithm exhibits superior efficiency for $N_q \geq 4$ and achieves approximately a threefold speedup for $N_q \geq 6$.



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Concluding Remarks

- **An efficient algorithm designed for robust control optimization in open quantum systems**
  - **Utilizes the Suzuki–Trotter expansion to accelerate system-dynamics evaluation**
  - **Outperforms standard GRAPE in optimization efficiency, particularly for high-dimensional quantum systems**

- **Future prospects**
  - **Incorporate broader classes of uncertainties, including multiplicative and time-dependent perturbations**
  - **Further enhance the optimization efficiency**
  - **Extend the approach to more complex control tasks**